

Nightingale

TECHNICAL BRIEF · FEEDBACK-INTEGRATED BUILD

Status: synthetic-data prototype — verified with explicit limits Baseline: main, 03 Sep 2026

The supplied clinic feedback numbers scenarios 1–16 and then adds a cross-cutting capability checklist. This brief treats that checklist as synthesis item 17; it does not invent a separately numbered source scenario.

1. Current build and integration model

Nightingale is one longitudinal care record. A real-world **Event** is the Timeline unit; parallel **Artifacts** preserve raw, AI-scribed, clinician, staff and patient-facing representations without overwriting one another. Every derived Highlight binds to the source Artifact version, exact Span, verbatim quote and quote hash.

CLIENT PROJECTIONS	SERVER AUTHORITY	DURABLE / EXTERNAL
Timeline · what happened Glance · what matters now Patient View · what to do	RBAC + clinic scope state + human gates ranking + provenance	SQLCipher/SQLite metadata audit one redacted LLM egress
Patient -> Event -> Artifact -> exact Span / version / hash		

The server owns scope, authorship, state transitions, safety, Task authority, publication and provenance. AI may assist language only. Formal Glance stays deterministic **base_only**; learned ranking is Shadow-only.

2. Assumptions re-evaluated

STILL STAND	NO LONGER STAND
• Event → Artifact → Span is the canonical model. • Raw source is immutable; human authority remains separate. • Server scope and patient-safe projection are mandatory. • Deterministic ranking precedes learned ranking. • The evidence is a synthetic, single-machine prototype.	• Email identity reaches every patient. • Redaction-before-model closes all privacy exits. • Batch ASR supports real-time alerts. • A generated link proves delivery. • Surfaced-only feedback is sufficient learning evidence. • Provider failure is always a returned error, not a hang.

3. Feedback traceability · scenarios 1–9

#	STATUS	FEEDBACK / BUILD INTEGRATION / FIRST FAILURE
1	DOES NOT	Phone-only identity. Email/password blocks a phone/WhatsApp-only patient. Server sessions are sound, but OTP, consented delivery and assisted offline access do not exist.
2	PARTIAL	Clinic isolation. RBAC + scoped loaders + FKs/triggers/indexes cover tested routes. SQLite has no RLS; a missed route remains the first leak risk.
3	PARTIAL	Logs and retention. Application fields are allowlisted/scrubbed and edge access logs are off. Host, crash, third-party and Provider retention remain unestablished.
4	SURVIVES	Redaction ordering. Recursive redaction precedes the sole LLM egress; restore and exact anchoring are local and fail closed. A future direct Provider call is the regression risk.
5	PARTIAL	Second clinic. Bootstrap, settings and idempotent CSV import need no schema rewrite. Organisation verification, provisioning, per-clinic billing/support and RLS remain.
6	PARTIAL	Trilingual consult. UTF-8, manual speakers, attestations and exact spans work. Local ASR missed two Malay phrases; fallback is mainly English-keyword and clinical validity is unproven.
7	DOES NOT	Minute-two allergy. Voice is post-consult batch. Streaming chunks, incremental risk detection, latency targets and alert acknowledgement/escalation are a separate product path.
8	PARTIAL	45-second hang. A 30-second total deadline cancels the request and returns labelled fallback/unavailable. Cancellation was proven locally, not against a deliberately hung live Provider.
9	SURVIVES	Provider 503. Raw-first ingestion plus labelled deterministic fallback remains useful and may safely produce zero Highlights. It is not equivalent to multilingual Provider understanding.

4. Feedback traceability · scenarios 10–17

#	STATUS	FEEDBACK / BUILD INTEGRATION / FIRST FAILURE
10	SURVIVES	Concurrent edits. CAS yields one winner and a visible 409; snapshots, diff, revert-as-new-version and audit explain it. No distributed certification or live presence.
11	PARTIAL	Delivery end-to-end. Exact-version portal receipts distinguish view/acknowledge. There is no sender, Provider receipt, retry, bounce or escalation for email/SMS/WhatsApp.
12	PARTIAL	Wrong patient copy. Clinician publish/correct/withdraw gates visibility and preserves history. External copies cannot be recalled; medication accuracy still depends on human review.
13	SURVIVES	Conflicting allergy. Both exact sources remain and needs_review outranks routine items; the system does not pick truth. The parser is bounded English logic.
14	PARTIAL	Importance meaning. It is a retrieval-order score, not risk probability. Factors, source and Coverage Review make it falsifiable; clinical calibration/outcome evidence is absent.
15	PARTIAL	Learning bias/fatigue. Complete decisions, scope, caps, protections, replay/freeze/rollback exist. No real labels or fatigue policy; Shadow only and formal Glance stays base_only.
16	SURVIVES	Edited source. Version + exact Span + quote hash resolves historical text or fails closed. No dependency graph or production archive certification.
17	PARTIAL	Cross-cutting synthesis. The concluding checklist is integrated where bounded. Streaming, diarization, noisy-clinic validation, medical-reference confirmation, regeneration and production learning remain.

Official numbered scenarios 1–16: **5 SURVIVE / 9 PARTIAL / 2 DOES NOT**. Cross-cutting synthesis item 17: **PARTIAL**.

5. What we tried, what failed, and where work stopped

DeepSeek protocol	The first Anthropic-format run spent all 2,000 output tokens on reasoning and returned no final content. The OpenAI-compatible final-content path later produced one strict synthetic summary with 4/4 anchors; this is not clinical validation.
Local ASR	Offline mechanics passed on supplied synthetic audio, but two Malay phrases were misrecognized and no diarization exists. We stopped at required human correction/attestation.
Tenant defense	SQLite cannot supply database RLS. Scoped loaders, triggers, FKs, indexes and fault injection reduce blast radius; production tenancy awaits a database/platform decision.
Observed learning	The compiler found zero eligible real-clinician pairs, so training correctly stopped. Frozen synthetic models remain Shadow-only and never serve Glance.
External boundaries	A local hanging server proved timeout cancellation; no live Provider hang was induced. Messaging, host/crash retention, streaming alerts and recall of sent content were not built or simulated.

6. Verification, evidence map, and foreseeable failures

BACKEND	FRONTEND	PROVENANCE / PROVIDER	PERFORMANCE
714 collected 712 passed 2 default ASR-input skips	Contract checks passed 112-module build passed	One synthetic DeepSeek run 4/4 exact anchors	Glance HTTP P95 5.115 ms 100 local warm samples

Scenario-linked automated evidence

The README maps each numbered scenario to its primary backend tests; dated evidence files preserve exact commands, observations and non-claims.

Foreseeable first production failures

Phone-only patient exclusion; ungoverned host/third-party retention; code-switching ASR error propagation; absence of in-consult alerts; undelivered external links; and over-trust in an uncalibrated retrieval score. Human gates, immutable raw sources, exact fail-closed provenance and deterministic serving limit—not eliminate—these risks.

Release boundary

This is a **synthetic-data prototype**, not evidence of production medical safety, multilingual clinical accuracy, public-host security, production multi-tenancy, real-time alert delivery, a clinically valid learned ranker, regulatory compliance or external message delivery.